Intelligent Mango Leaf Disease Detection and Classification Using Transfer Learning and Explainable Artificial Intelligence

Abstract

Mango is a vital fruit crop, both economically and nutritionally. Its productivity plays an important role in food security and the economy of many developing countries. However, mango trees are susceptible to a variety of leaf diseases that can severely affect both yield and quality. The most marked among these diseases are Anthracnose, Bacterial Canker, Cutting weevil, Die Back, Gall midge, Powdery Mildew and Sooty mold. Early and accurate detection of these diseases is crucial for farmers and recent advancements in modern technology now enable automated identification. A deep learning-based disease detection system has been developed using a dataset of 12,729 mango leaf images categorized into eight disease classes. The performance of several transfer learning models—MobileNetV2, VGG16, EfficientNetB0 and ResNet50 has been analyzed. Furthermore, the ResNet50 model has been enhanced with Grad-CAM-based explainable AI to improve both interpretability and performance. Experimental results demonstrate that the ResNet50 + Explainable AI model achieves the highest accuracy of 99.36%, outperforming the other models. This explainable model not only accurately identifies diseases but also highlights the specific leaf regions that influenced its predictions through Grad-CAM heatmaps, thereby increasing user trust. The proposed method is efficient, cost-effective, suitable for real-time field deployment and offers a practical disease monitoring solution for farmers.

DOI

https://doi.org/10.1109/RAAICON69033.2025.11502136